

CSE221 Lesson Plan, Fall 2025 Mid-term: Nov 16, Sunday Final: Jan 11, 2026, Sunday		Course Outline: https://docs.google.com/document/d/1I0ub-Dn-8wnYDPwRU2ce8Nrtal
Class #	Topics	Materials
0	Ice breaking	https://m.media-amazon.com/images/I/51oXcGxCoFL._AC_SX679_.jpg
1	1. Introduction: Algorithms for accuracy, efficiency (space, time); Algorithms in real world 2. Time Complexity (recap): RAM Model, Loop iterations vs execution time, Growth Functions, best-worst scenario	Iterative time complexity: https://drive.google.com/file/d/1rTgD7Vzsvd4jWFr9taPqrddtOHlHXF
2	Searching algorithms 1. Binary: Logical explanation on the logarithmic time complexity, fitness/feedback function, finding upper - lower bound 2. Ternary: Concept, Advantages over binary search 3. Balancing between no. of search steps and cost of each step: why quaternary search is not that popular	Linear: https://drive.google.com/file/d/1vAJ2EbjdNmV6_ZgbEcDwDjJ3RnwPs7qp/view Binary: https://drive.google.com/file/d/1P6QjoYShah1YIFzdgNCRbnXd379BiuQb/view Ternary: https://drive.google.com/file/d/1tP_ebyE3uHpGto6NlCdF5-VIzUsgoyOe/view
3	Sorting algorithms 1. Case analysis on N^2 sorting algorithms: Bubble, Insertion/Selection; # of iterations, # of swaps on best/worst scenario 2. Linear (case specific) time sorting: Count sort, a good example of why nested iterations is not always multiplied for time complexity	Animated simulations from: https://visualgo.net/en/sorting
4	Sorting contd. 1. $N \lg N$ sorting: Merge, Quick (fixed and random pivots)	Merge: https://drive.google.com/file/d/1JUGTeiVhRnwaMKVcvwurikKGymIq-U25/view?usp=s Quick: https://drive.google.com/file/d/1vbcB0tf5jklJBLzMbniFr3FZlvF3nbO/view?usp=sharing
5	Quiz + Lagged discussion (if any)	
6	Asymptotic Notations: upper, lower, tight bound complexity	Iterative time complexity: https://drive.google.com/file/d/1rTgD7Vzsvd4jWFr9taPqrddtOHlHXF
7	Divide and Conquer (DnC) algorithms 1. Look back to merge, quick sort 2. Max-sum subarray: $O(N \lg N)$ [can also be solved in $O(N^3)$, $O(N^2)$, $O(N)$]	Max-sum subarray: https://docs.google.com/presentation/d/1rIC6OXh7W5ELsxkfGbnYWJRS
8	DnC contd.: Karatsuba multiplication	Karatsuba: https://docs.google.com/presentation/d/1xcTFil_wxMCVomD54YsKQxytc6VfNaf3
9	Time complexity of recursive/DnC algorithms: recursion tree, substitution method, master theorem	Recursive Time complexity: https://drive.google.com/file/d/12cR_JBR8wZg0NDAIL-wzL6fUsZ
10	Quiz + Review	
Mid Semester Examination		
11	BFS: traversal (review), shortest path on unweighted graph, Bicolorable/Bipartite graph detection, cycle detection (odd or even length)	bfs dfs: https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWGk6V91q/view?usp=s
12	DFS: traversal (review), Edge classification, cycle detection	bfs dfs: https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWGk6V91q/view?usp=s
13	What is DAG, Topological ordering, Strongly Connected Components (SCC, Kosaraju's algo)	top sort: https://drive.google.com/file/d/1U-PVynHjD-PytCeHvcqtCsCaEE-iL-qc/view?usp=sh SCC: https://drive.google.com/file/d/1fckjavRyW4yCAi2zmAjsF3pKhlsr3FJb/view?usp=sharing
14	Shortest path algo: What is SSSP and APSP, Dijkstra's algo, <i>Bellman-Ford Algo</i> (Negative-weight edges in-play)	MST+Dijkstra: https://drive.google.com/file/d/1zNpTudRm--5wPHKrownXyosZQ1ta69F3/view
15	MST: Kruskal's (+DSU)	MST+Dijkstra: https://drive.google.com/file/d/1zNpTudRm--5wPHKrownXyosZQ1ta69F3/view
16	Quiz + Lagged discussion (if any)	
17	Greedy: Huffman coding, <i>Interval scheduling (covered in lab before mid)</i>	Greedy Basics, time scheduling: https://drive.google.com/file/d/1U-LLjWQHP-RdARYV8vOYI Huffman: https://drive.google.com/file/d/1-_oOWBKQARdaWgZvLTUFIv5tCYd5CqZ4/view?u
18	Backtracking Algorithms: 8 queen Problem	
19	Backtracking Algorithms: 15 puzzle Problem	
20	DP-1: Fractional vs 0-1 Knapsack	fractional knapsack: https://drive.google.com/file/d/1U-LLjWQHP-RdARYV8vOYIMkKglZBFZc DP Basics, 0-1 knapsack: https://drive.google.com/file/d/1tdh2T7G4L4uMVuLnxxvqHhJqVxku
21	DP-2: LCS, Recurrence relation, Properties: overlapping subprob and opt. substructure	LCS: https://drive.google.com/file/d/1YJonhR6glZoGMLk0G6323Mjytohllvn6/view?usp=shar
22	Quiz + Review	
Final Examination		
Marks Distribution		Quiz: 20 (best 3 out of 4, or as instructed by course teacher) Assignment: 05 (4 assignments) Mid Semester Exam: 20 Final Exam: 30 Lab: 25

